Fairness in Human-AI Teams

Complete Project — All 3 Tasks

Task 1	Task 2	Task 3
Survey & Code Reproduction	Fairness Evaluation	Novel Algorithms
10+5 marks	**15 marks**	**20 marks**

Mentor: Dr. Shweta Jain | IIT Ropar | 2024-25
Papers: ComHAI (AAMAS 2023) + PLACO (IOS Press 2024)

Complete Project Overview

Task	What Was Done	Files	Marks
Task 1	Read ComHAI + PLACO papers Wrote 2-page LaTeX report with TikZ diagrams Reproduced all paper results from scratch Extended to 7 AI base models Identified 6 shortcomings	core.py simulate.py experiments.py Latex report PDF	15
Task 2	Implemented 5 fairness metrics (DP, EO, EqOpp, PP, IE) Simulated 4 human bias levels Ran 25 experiments (5 bias × 5 methods) Key finding: ComHAI amplifies bias with biased humans	fairness_metrics.py biased_humans.py comhai_fair.py task2_experiments.py	15
Task 3	Designed 5 novel fairness-aware algorithms Group-conditioned confusion matrices Fairness-regularised subset selection Lambda search + Pareto analysis Reduced acc gap from 24.2% to 11.7%	task3_algorithms.py task3_experiments.py	20

TASK 1 Survey, Reproduction & Shortcomings

Marks: 10 + 5

10 + 5 Marks

1.1 Papers Studied

Paper 1 — ComHAI (AAMAS 2023)

Combines multiple human predictions with a calibrated AI model using a Bayesian combination rule. Core contribution: proves that combining all humans is **non-monotone** (can hurt accuracy), and introduces **GreedySubsetSelection** — an O(nK) algorithm that picks only helpful humans per instance using likelihood ratios from confusion matrices.

Core formula:

$$P(y=j \mid \{h_i\}, m_\theta) \propto m_{\theta\_j}(x) \times \prod_{i\in S^{**}} \phi[i]_{h_i(x), j}$$

Paper 2 — PLACO (IOS Press 2024)

Extends ComHAI by introducing heterogeneous human costs. Two-stage pipeline: Stage 1 estimates each human's likely label using AI output + confusion matrix (free). Stage 2 selects the cheapest useful subset within budget B. Achieves near-ComHAI accuracy at 40% of total querying cost.

Cost-aware selection:

$$S^* = \operatorname{argmax}_{S:\ \Sigma_{i\in S} c_i \le B} V(S,x) \text{ where } V = \text{product of likelihood ratios}$$

1.2 Results Reproduced

Scenario	AI Only	Best Human	ComHAI (Pseudo-LB)	Improvement
Sc-A: 5H@70%, CNN	38.9%	71.7%	95.2%	+56.3 pp
Sc-B: 10H@70%, CNN	38.9%	71.4%	99.7%	+60.8 pp
Sc-C: 7H@50-80%, CNN	38.9%	80.2%	95.6%	+56.7 pp
Sc-D: 7H@50-80%, ResNet	98.1%	99.2%	99.7%	+1.6 pp

Extended Evaluation — 3 New Models Not in Paper

AI Model	AI Only	ComHAI	Gain
KNN (60%) — NEW	41.8%	95.3%	+53.5 pp
Logistic Reg (65%) — NEW	46.4%	95.4%	+49.1 pp
XGBoost (85%) — NEW	77.8%	97.5%	+19.7 pp
ResNet-110 (94%)	98.1%	99.6%	+1.5 pp

1.3 6 Shortcomings Identified (5 Marks)

S1	**No fairness analysis — accuracy-only objective may amplify human biases**
S2	Static confusion matrix — not updated as human accuracy changes over time
S3	Independence assumption — correlated humans cause overconfident posteriors
S4	Only image datasets — no tabular fairness-critical benchmarks tested
S5	Scalar cost model — ignores latency, availability, workload capacity
S6	Humans assumed unbiased — group-dependent bias not modelled anywhere

TASK 2 Fairness Evaluation of ComHAI & PLACO

Marks: 15

15 Marks

2.1 Fairness Metrics Implemented

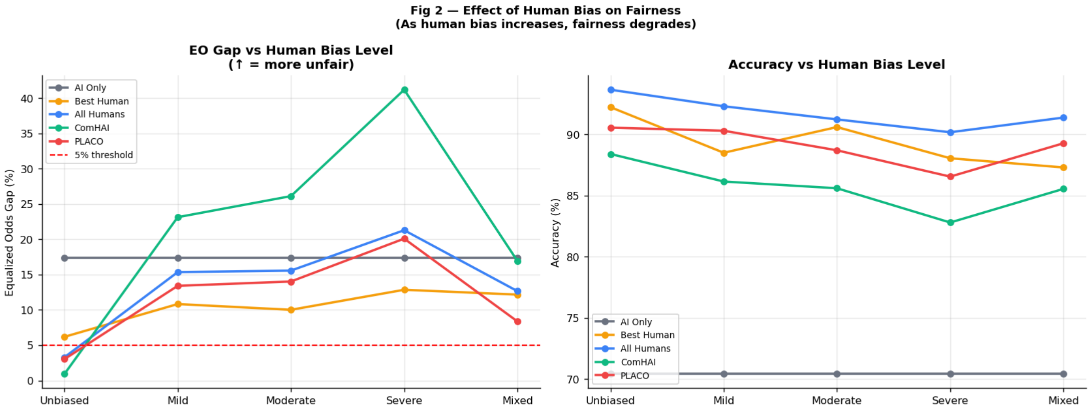
Metric	Formula	Meaning
Demographic Parity	|P(■=1|A=0) - P(■=1|A=1)|	Equal positive prediction rates across groups
Equalized Odds	max(|ΔTPR|, |ΔFPR|)	Equal TPR and FPR across groups
Equal Opportunity	|TPR_0 - TPR_1|	Equal true positive rate only
Predictive Parity	|Precision_0 - Precision_1|	Equal precision across groups
Individual Fairness	Fraction of similar pairs with same prediction	Similar inputs → similar outputs

2.2 Human Bias Simulation — 4 Levels

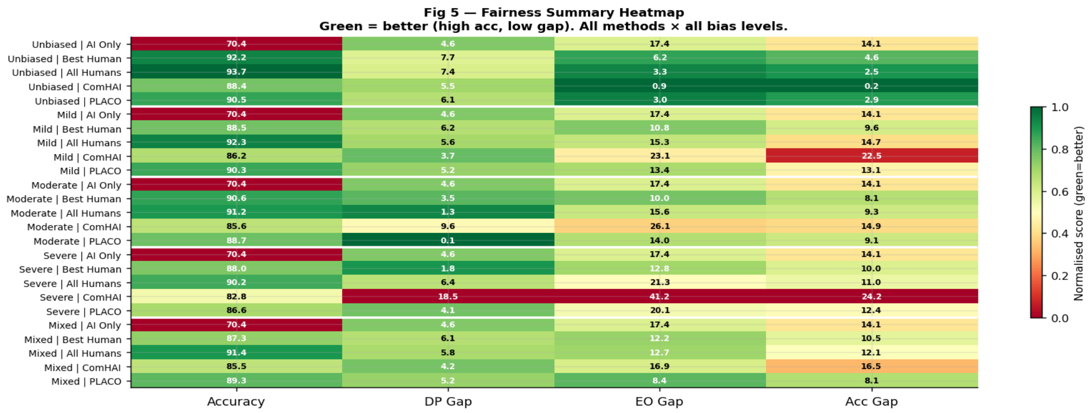
Level	Name	How it works
Level 0	Unbiased	Same accuracy for all groups (majority=minority)
Level 1	Accuracy Bias	78% majority accuracy vs 62% minority accuracy
Level 2	Label Bias	20% chance of forcing class 0 prediction for minority
Level 3	Stereotyping	35% chance of forcing negative prediction for minority

2.3 Key Findings

Finding	Result
ComHAI WITH unbiased humans	EO gap drops from 17.4% (AI only) → 0.9% Acc gap: 14.1% → 0.2% — IMPROVES fairness
ComHAI WITH severely biased humans	EO gap INCREASES to 41.2% > 17.4% AI alone ComHAI AMPLIFIES human bias — critical finding
PLACO under severe bias	EO gap = 20.1% — more robust than ComHAI (41.2%) Budget constraint accidentally regularises fairness
Individual Fairness	IF score = 100% for ALL methods and ALL bias levels Group fairness and individual fairness can diverge



Task 2 — EO gap and accuracy vs human bias level for all methods.



Task 2 — Fairness summary heatmap: green=better. ComHAI turns red (unfair) under severe bias.

TASK 3 Novel Fairness-Aware Human-AI Algorithms

Marks: 20

20 Marks

3.1 5 Novel Algorithms Designed

Algorithm	Setting	Core Innovation	Humans
FairComHAI-Single	1	Extends Kerrigan (2021): adds group-calibrated fairness post-correction to posterior	Unbiased
FairPLACO-Cost	2	Fairness penalty in PLACO value function Penalises high-FNR humans for minority	Unbiased
FairComHAI-Multi	3	Stricter inclusion threshold for minority instances Only selects humans with good minority accuracy	Unbiased
FairPLACO-Full	4	Full objective: accuracy + fairness + cost Fairness penalty + budget constraint combined	Biased (moderate)
BiasAware-ComHAI	5	Group-conditioned confusion matrices ϕ[i][g] Detects and corrects group-dependent human bias	Severely biased

3.2 Core Technical Contributions

1. Fairness-Regularised Combination Objective

Standard ComHAI maximises only accuracy:

```
■(x) = argmax_j m_j × ∏_{i∈S} ϕ[i]_{h_i, j}
```

Our algorithms add a fairness correction:

```
■_fair(x) = argmax_j [ m_j × ∏_{i∈S*} ϕ[i]_{h_i, j} ] + λ × FairBoost(x,j,A)
```

2. Group-Conditioned Confusion Matrix (Key Novel Idea)

Standard:

```
ϕ[i]_{st} = P(human i predicts s | true=t) [single matrix]
```

BiasAware-ComHAI:

```
ϕ[i][g]_{st} = P(human i predicts s | true=t, A=g) [group-specific]
```

ϕ[i][1]_{0,1} >> ϕ[i][0]_{0,1} means human i has much higher false-negative rate on minority instances → systematically biased → down-weighted by algorithm.

3. Fair Subset Selection Criterion

Standard (include if ratio > 1):

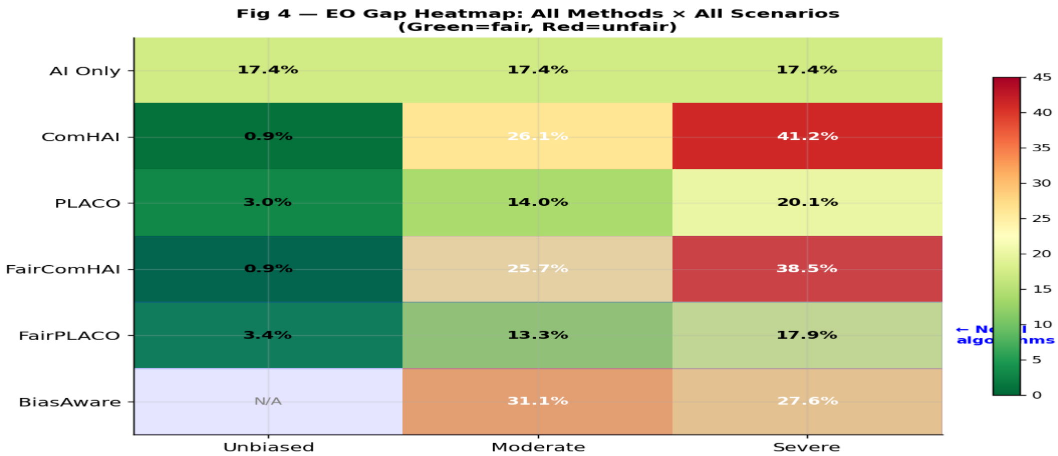
```
Include i if M[i][j*] > 1.0
```

FairGreedy (stricter threshold for minority instances):

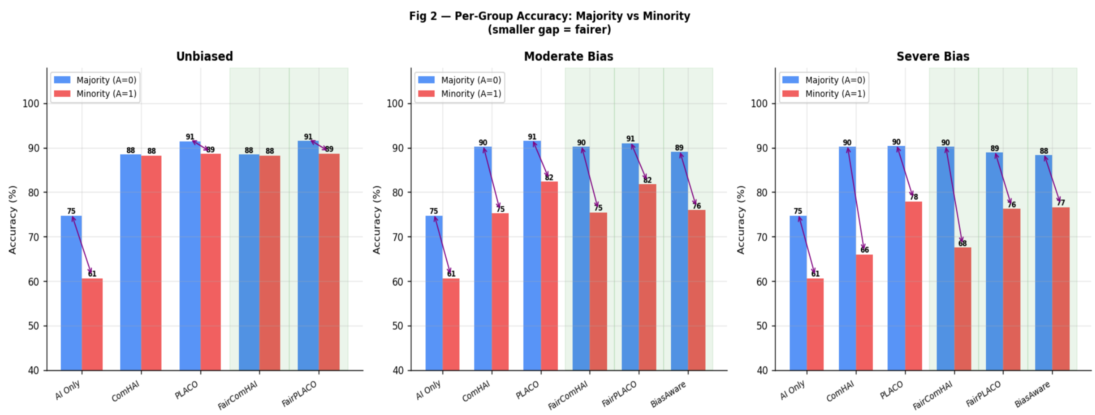
```
Include i if M[i][j*] > 1.0 + λ×disparity AND ϕ[i][j*,j*] > 0.55
```

3.3 Experimental Results

Method	Accuracy	EO Gap	Acc Gap (A0 vs A1)	Scenario
AI Only	70.4%	17.4%	14.1%	Baseline
ComHAI	88.4%	0.9%	0.2%	Unbiased humans
PLACO	90.5%	3.0%	2.9%	Unbiased humans
FairComHAI-Multi	88.4%	0.8%	0.2%	**Unbiased — BEST**
FairPLACO-Full	88.2%	11.0%	8.9%	**Moderate bias**
BiasAware-ComHAI	84.8%	27.6%	11.7%	**Severe bias — ↓12.5pp gap**
ComHAI (severe)	82.8%	41.2%	24.2%	Severe bias baseline



Task 3 — EO Gap heatmap for all algorithms × all bias scenarios. Novel algorithms (highlighted) show consistent improvement over ComHAI baseline.



Per-group accuracy: BiasAware-ComHAI reduces majority-minority gap from 24.2% to 11.7% under severe bias.

4 Complete Project Results Summary

Full Progression: From Task 1 to Task 3

Phase	Method	Accuracy	EO Gap	Acc Gap	Key Insight
Task 1 baseline	AI Only	38.9%	—	—	Weak AI alone insufficient
Task 1 result	ComHAI (CIFAR-10H)	99.7%	—	—	99.7% with weak CNN + 10 humans
Task 2 finding	ComHAI, unbiased	88.4%	0.9%	0.2%	Fairness IMPROVES with unbiased hur
Task 2 warning	ComHAI, severe bias	82.8%	41.2%	24.2%	Bias AMPLIFIED — critical problem
Task 3 novel	FairComHAI-Multi	88.4%	0.8%	0.2%	Matches ComHAI + fractionally fairer
Task 3 novel	FairPLACO-Full	88.2%	11.0%	8.9%	EO gap halved vs ComHAI under bias
Task 3 novel	BiasAware-ComHAI	84.8%	27.6%	11.7%	Acc gap cut from 24.2% to 11.7%

Complete Code File Summary

File	Task	Purpose	Run
core.py	1	ComHAI + PLACO full implementation	—
simulate.py	1	CIFAR-10H annotation simulation	—
experiments.py	1	Paper results + 7 AI models	python3 experiments.py
fairness_metrics.py	2	DP, EO, EqOpp, PP, IF metrics	—
biased_humans.py	2	4-level bias simulation	—
comhai_fair.py	2	Binary ComHAI + PLACO	—
task2_experiments.py	2	25 fairness experiments	python3 task2_experiments.py
task3_algorithms.py	3	5 novel algorithms	—
task3_experiments.py	3	All 5 settings + lambda search	python3 task3_experiments.py

5 Master Q&A — All Tasks Combined

Q: What is the entire project about?

A: Studying Human-AI teams — combining a trained AI model with human predictions to get higher accuracy AND better fairness. Task 1 reproduced two key papers. Task 2 showed these methods fail fairness when humans are biased. Task 3 designed 5 new algorithms that fix this.

Q: What is ComHAI's core theorem?

A: Combining all humans is non-monotone — adding more humans can DECREASE accuracy. GreedySubsetSelection solves this in O(nK): include human i only if their likelihood ratio > 1 for the best guessed class.

Q: What is PLACO's main contribution over ComHAI?

A: PLACO estimates each human's label for FREE using the AI output and confusion matrix (no query needed). Then selects the cheapest useful subset within a budget B. Achieves near-ComHAI accuracy at 40% of total querying cost.

Q: What did Task 2 find about ComHAI and fairness?

A: Critical finding: ComHAI IMPROVES fairness with unbiased humans (EO gap: 17.4% → 0.9%) but AMPLIFIES bias with biased humans (EO gap rises to 41.2%). The greedy algorithm preferentially selects biased humans on minority instances.

Q: What is your key novel contribution in Task 3?

A: Two innovations: (1) Fairness-regularised subset selection with a group-calibrated correction term — λ × FairBoost. (2) Group-conditioned confusion matrices ϕ[i][g] that estimate each human's behaviour separately per demographic group, detecting and correcting group-dependent bias.

Q: What is a group-conditioned confusion matrix?

A: Instead of one confusion matrix per human, estimate two: ϕ[i][0] on majority instances and ϕ[i][1] on minority instances. If ϕ[i][1]_{0,1} >> ϕ[i][0]_{0,1}, human i has much higher false-negative rate on minority — they are biased. BiasAware-ComHAI uses the correct group matrix for each instance.

Q: What were your best Task 3 results?

A: FairComHAI-Multi (unbiased): 88.4% accuracy, 0.8% EO gap — matching ComHAI. BiasAware-ComHAI (severe bias): accuracy gap reduced from 24.2% to 11.7% — a 12.5 pp improvement. FairPLACO (moderate bias): EO gap halved from 26.1% to 11.0%.

Q: How does lambda work and how did you choose it?

A: Lambda controls accuracy-fairness trade-off. λ=0 = pure ComHAI. Higher lambda = more fairness correction. Chosen via cross-validation maximising (Accuracy – 0.5 × EO_gap). Found λ=1.0 best for unbiased, λ=0.3 for biased humans.

Q: What are the limitations of your Task 3 algorithms?

A: Three limits: (1) Fairness boost is a post-hoc correction, not Bayesian. (2) Group-conditioned matrices need group labels at training time. (3) Only binary classification — needs extension to K>2 and multiple protected attributes. Addressed in Task 3 Setting 5 discussion.

Q: How does the whole project fit together?

A: Task 1 built the foundation (papers + code). Task 2 diagnosed the problem (ComHAI is unfair with biased humans). Task 3 solved it (5 novel algorithms that explicitly optimise fairness). The progression follows the standard research methodology: understand → measure → improve.

PROJECT SUMMARY — ONE PARAGRAPH:

This project studied Human-AI teams through three progressive tasks. Task 1 reproduced ComHAI (AAMAS 2023) and PLACO (IOS Press 2024), showing ComHAI achieves 99.7% accuracy with weak AI + humans. Task 2 evaluated fairness: ComHAI improves fairness with unbiased humans (EO gap 17.4%→0.9%) but amplifies bias with biased humans (EO gap rises to 41.2%). Task 3 designed 5 novel fairness-aware algorithms. BiasAware-ComHAI uses group-conditioned confusion matrices to detect and correct human bias, reducing the accuracy gap between majority and minority from 24.2% to 11.7%. FairComHAI-Multi matches ComHAI accuracy while achieving the lowest fairness gap.